

Agile Methodology: A Flexible Approach In Modern Project Management

Lee Yin Shen, Choh Jing Yi
Department of Computer Science
on Data Engineering
Faculty of Computing
Universiti Teknologi Malaysia
Johor, Malaysia
leeshen@graduate.utm.my
choh@graduate.utm.my

Abstract—Agile methodology is a modern approach for project management which focus on flexibility, teamwork, and continuous improvement. This paper explains the Agile principles and introduces popular frameworks like Scrum and Kanban. Moreover, this paper highlights the benefits of Agile which include enhance project adaptability and improve customer satisfaction. This paper also discusses the common challenges and solutions that will be facing by an Agile Team, such as communication issues. Lastly, by comparing Agile to traditional methods, this paper show how Agile can bring better project outcomes in this dynamic environment.

Keywords—Agile, project management, flexibility Scrum, Kanban, adaptability, customer satisfaction

I. INTRODUCTION (HEADING 1)

Software development techniques have changed significantly in the current fast-paced technological world to meet the demands of complex and adaptable environments. The Agile methodology in project management is a structured approach that divides projects into manageable parts, prioritising continuous improvement. It is a cyclical process encompassing planning, execution, and evaluation. [1]. The Agile methodology originated with the Agile Manifesto in 2001 highlighting the requirement of high-quality software through continuous development and feedback loops, promoting for close collaboration among clients. [2]. In contrast to conventional models that depend on inflexible and linear procedures, Agile methodology embraces a more dynamic methodology which more prioritising iterative development, flexible planning and ongoing enhancement. The principles and frameworks of Agile such as Scrum and Kanban have transformed software engineering methods and impacted other fields. This paper examines Agile principles, common Agile frameworks, benefits of Agile methodology, challenges of Agile methodology, and a comparison of Agile versus traditional methodology.

II. AGILE PRINCIPLES

- A. Fulfil customer satisfaction by providing valuable software quickly and consistently.

- The first principle centres on ensuring customer satisfaction through early and continuous delivery of valuable software. The Agile mindset prioritizes product design based on the customer's viewpoint rather than adhering to a "we know what is best" paradigm [3].
- B. Embrace changing requirements during the development process even in the later stages of the project timeline.
- This principle reflects Agile's adaptable nature, enabling project teams to respond to changing customer requirements or market demands without the limitations of a fixed plan.
- C. Deliver working versions frequently.
- The Agile approach emphasizes subdividing projects into smaller segments or sprints, delivering a working version of the product at the end of each sprint. For example, scrum structures projects into multiple short sprints, which are time-constrained events lasting one month or fewer, resulting in a concrete delivery and a retrospective evaluation [3].
- D. Collaboration between business stakeholders and developers throughout the project.
- The Agile Manifesto asserts that business personnel and developers should collaborate regularly during the project. It is imperative that stakeholders engage in communication and collaboration to eliminate uncertainty and guarantee a successful project conclusion [3].
- E. Builds projects around motivated individuals
- Agile principle highlights the significance of self-organizing teams that possess autonomy in achieving project objectives. Agile leaders understand that individuals will perform optimally and yield high-quality results when provided with an appropriate atmosphere, adequate support, and mutual trust. Enhancing the speed and efficiency of your initiatives demands assembling an appropriate team, providing essential resources, and avoiding micromanagement [3].
- F. Face-to-face conversation for sharing information, within a development team [4].

- This principle recognizes the limitations of written documentation or asynchronous communication in articulating intricate concepts, thus promoting direct interactions to improve clarity and engagement.
- G. Continue to focus on quality design and technical proficiency.
- Agile prioritizes ongoing testing and monitoring to identify and rectify issues promptly. Traditional project management methodologies, like Waterfall, adopt a linear approach, with testing completed towards the final stage of the project [3] .
- H. Use self-organizing teams.
- Agile prioritizes self-organizing teams in cultivating innovative and high-quality outcomes, as these teams are authorized to make independent decisions, utilizing their collective skills to address complicated challenges. A Monster survey indicated that 73% of employees regard micromanagement as the foremost workplace "red flag," while 46% cite it as a factor prompting them to resign [5].
- I. Regular reflect and review through retrospective sessions
- This enables your team to remain flexible and seek opportunities to enhance their procedures and performance. This mentality facilitates the attainment of more principles, including discovering more efficient methods and embracing transformation.

III. COMMON AGILE FRAMEWORKS

Agile frameworks are iterative workflow phases that systematize communication, planning, management, implementation, and testing to get a given project outcome within a defined timeframe. These frameworks offer distinct structures and standards that facilitate the implementation of Agile principles across diverse projects and organisational contexts.

A. Scrum

Scrum is a management methodology that enables teams to self-organize and achieve a shared objective [6]. It establishes a framework of meetings, instruments, and responsibilities for effective project execution. The Scrum framework includes defined roles (Product Owner, Scrum Leader, and Scrum Development Team), ceremonies (Sprint Planning, Daily Standup, Sprint Review, and Retrospective), and artifacts (Product Backlog, Sprint Backlog, and Increment) that structure and facilitate team collaboration[6].

B. Kanban

Kanban is a popular framework used to implement Agile and DevOps software development[7]. It is a lean workflow management method designed to improve efficiency, minimize waste, and enhance overall

productivity [8]. It requires real-time communication of capacity and full transparency of work. Work items are represented visually on a kanban board, allowing team members to see the state of every piece of work at any time.[7]

Key element of Kanban board [8]:

- Visual the Workflow
The kanban system utilises a kanban board, serving as a visual representation of the team's work items and their corresponding stages of completion.
- Measure and Manage Flow
Metrics including cycle time, lead time, and throughput are monitored to identify supply chain bottlenecks and enhancement opportunities.

C. Extreme Programming

Extreme programming is an Agile project management methodology that targets speed and simplicity with short development cycles and less documentation [9] . Extreme Programming is the most specific of the agile frameworks regarding appropriate engineering practices for software development.

Key Practice of Extreme Programming:

- Pair Programming
Pair programming is an Agile software development technique originating from Extreme programming in which two developers' team together on one computer. The two people work together to design, code and test user stories.[10]
- Test-Driven Development (TDD)
TDD is a methodology in software development that focuses on an iterative development cycle where the emphasis is placed on writing test cases before the actual feature or function is written.

IV. BENEFITS OF AGILE METHODOLOGY

Agile methodology brings numerous benefits in term of increasing project outcomes, improving team productivity and enhancing customer satisfaction.

A. Improve Project Adaptability

Agile methodology has become a popular choice for improving project adaptability, especially in the dynamic environments where the project requirements will change frequently. By breaking the projects into smaller subprojects which are more manageable, it allows a team to shift their strategies quickly, without interrupt the main flow of the conducting project [12].

Moreover, Agile methodology enhance project adaptability by providing continuous delivery. Continuous delivery enables a team to make changes frequently and effectively base on the continuous feedback obtained from the customers [13], [14]. This enables a team refine their project

features in real time, which can ensure their project outcome always stays relevant with the dynamic environment.

B. Increase Customer Involvement and Satisfaction

Agile methodology highly focuses on customer involvement and satisfaction by collecting real-time feedback throughout the whole development process. Different with traditional method which only collect customer feedback during testing section, Agile involves stakeholders at every process iteration. Basically, an Agile team will hold daily stand-up meeting with customers and demonstrate the current system flow or product, which enable to obtain real-time feedback from customers [14]. This engagement allows an Agile team always align closely with the customer needs and making adjustment on their project in order to ensure the final project outcome match the customer expectations [13].

V. CHALLENGES AND SOLUTION

A. Managing Scope Creep

Since Agile facing a dynamic environment, managing scope creep become a crucial challenge. Although Agile encourage adaptability, the frequent change or update of the requirements will make the team unable to manage their project [16], finally lead to missed deadline. Moreover, the misunderstanding during the meeting with stakeholders will cause ambiguity on the project requirements [16], [17], which can make a team struggle to meet their initial goals while accommodating new elements and finally affect the outcome quality.

To address the scope creep, careful planning and transparent communication should be equipped in an Agile team. ScrumMasterd [17], Backlog Grooming sessions help to evaluate new requests from customers to ensure the project align with project goals. Refer to [16], Jira tool is used to track all requirement changes, and make it easier to monitor the scope adjustments. Roadmunk [18] mentions that SAFe PI planning sessions important to align stakeholders on priorities and help a team to prevent scope creep effectively.

B. Communication Gaps Between Teams and Stakeholders

Since an Agile team consists of people from different country, thus each members have different set of norms, expectations, values and work culture [19]. In addition, as a method which focus on real-time collaboration, the difference of time zones for each member causes difficulty to synchronize stand-up meetings and having face-to-face interaction, which cause information delays and disengagement of members [19]. Especially if a team conducting some urgent meeting, the absence of some team members will cause misalignment of project.

To address this challenge, an Agile team can utilize Application Lifecycle Management (ALM) tools like Jira and Trello [19]. These tools provide a centralized platform for team collaboration which centralizing all project updates and documentations. This can enable team members from

different time zone obtain the latest information and stay aligned with the latest progress [19]. Furthermore, video conferencing tools like Zoom or Google Meet should be used for important meetings to enhance personal interaction and reduce misunderstandings [19], [20].

VI. AGILE VS. TRADITIONAL METHODOLOGY

A. Project Structure and Planning

- Traditional methodology like Waterfall using a linear and sequential approach with distinct stages, which include requirement identify, analysis, design, implementation, testing and maintenance [22], [23]. Each phase will be begun only if the phases before it completed [22]. Moreover, the requirements in Waterfall are fixed and clear, and will be clearly defined before initiating a project [23].
- While for Agile method, it using iterative and incremental development approach which break a project into smaller and more manageable subprojects, and allow continuous adjustment to frequent update and improve the project [12]. Agile will continuously receive feedback from users which enable the project become more adaptive and flexible in dynamic environment [23].

B. Team Structure and Communication

- For traditional method like Waterfall Model, it relies on hierarchical team structure [23] which clearly distributes well-defined roles to each member. In term of communication style, traditional projects rely on formal communication which focus on written documentation like status report and email [24]. Moreover, the communication between each team is limited, normally occur when transferring phases.
- In agile team, there is no fixed hierarchy structure, all the roles may shift on count of dynamic environment. Instead, Agile teams are self-organizing and cross functional [23], which means all team members have ability to manage multiple tasks and roles and work closely together throughout the whole project. This structure allows a team to adapt with changing requirements more effectively. In term of communication, agile focus on real-time stand-up meeting and face-to-face meeting if possible [14]. This communication style enables team members to stay aligned with the latest information [14].

VII. CONCLUSION

In conclusion, the Agile methodology breaks tasks into small and flexible steps which allow teams to quickly adapt to changing requirements while delivering value incrementally [25]. This iterative method encourages close collaboration with clients, making sure it always meets their requirements, resulting in higher satisfaction. The ability to quickly adapt to new knowledge and market developments has made it popular across several industries, as it allows

businesses to sustain competitiveness by delivering higher quality goods faster. While Agile offers significant advantages over traditional methodologies, such as improved flexibility and team autonomy, it also presents challenges, including communication gaps in distributed teams and managing frequent requirement changes. Agile's ideas of adaptability, cooperation, and incremental advancement offer a strong foundation for modern software development and project management, leading to improved results and enhanced client satisfaction.

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